TIC 440817847

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

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BLS EPHEMERIS
  Period       3.338136 d  (ALIASED, see below)
    -> corrected period: 6.676272 d
  Epoch (t0)   1599.3242 BTJD
  Duration     0.2440 d
  Depth        2.790 %  (27,903 ppm)
  SNR          91.8

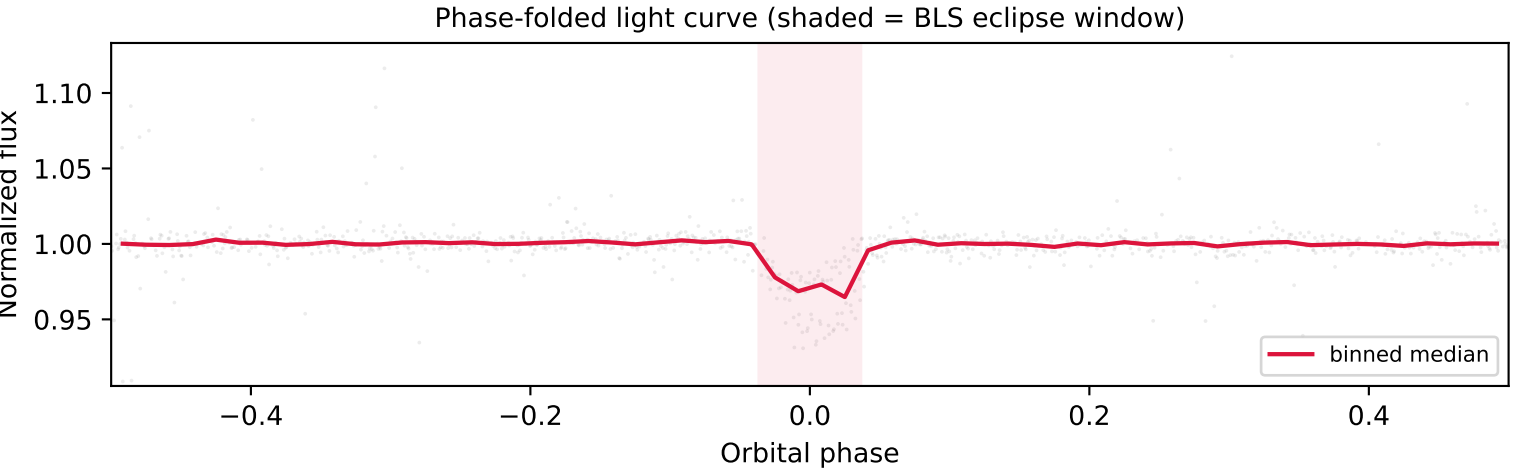
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     8  (gate >= 3)
  Robust depth ratio    0.93  (gate >= 0.50)
  Robust depth sigma    45.1  (gate >= 7.0)
  Odd/even z-score      6.44  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

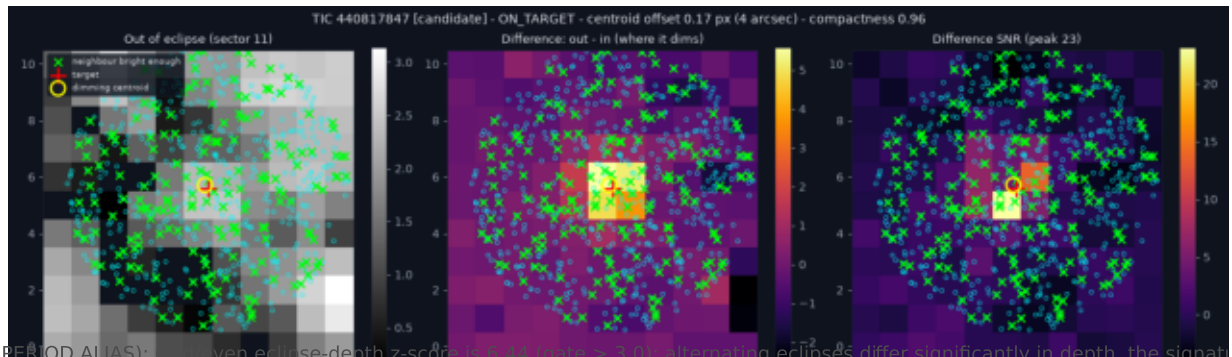
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 11    light-curve dip in this sector: 46.4 sigma    t0 refined by 0.0075 phase
  Difference-image peak SNR   23.18   (gate >= 5.0)      Compactness   0.96   (gate >= 0.50)
  Dimming centroid offset     0.17 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 440809684
    0.20 px away, dTmag +2.46 - farther from the dimming than the target
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Caveat (PERIOD ALIAS): ... even eclipse-depth z-score is 6.44 (gate > 3.0); alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 6.676272 d, in any write-up.
Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.029 px (dimming centroid 0.168 px from target vs 0.196 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.